Minkyu Kim

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Research and Engineering Projects

Reusable Agent Guidance for Robot Manipulation Aug 2026 – Present
MIPAL, Seoul National University
Advised by Prof. Nojun Kwak

- Contributed to method design discussions for transferring robot intervention experience between agents through a reusable playbook refined with execution feedback.
- Supported experiments on physical robots to evaluate the shared system, which improved manipulation success from 37.3% to 64.0% without updating model parameters.

Adaptive Execution Control for Robot Manipulation May 2026 – Present
MIPAL, Seoul National University
Advised by Prof. Nojun Kwak

- Built and deployed a learned execution controller on a Franka Panda, integrating policy fine-tuning, rollout collection, and verifier training. Across three manipulation tasks, increased success from 41.1% to 74.4% and reduced replans from 36.54 to 12.87 per episode against the best fixed-horizon baseline.
- Reduced pi0 inference calls on LIBERO by 64% while raising success from 92.1% to 94.3% against a fixed five-step baseline by predicting when to replan. Benchmarked pi0 and pi0.5 across LIBERO and RoboTwin 2.0.

Task-Completion Guidance for Robot World Models Oct 2025 – Sept 2026
MIPAL, Seoul National University
Advised by Prof. Nojun Kwak

- Co-designed inference-time guidance to help robots complete missing manipulation steps instead of repeatedly generating plausible but incomplete motions. Developed the mathematical analysis behind strengthening task-relevant language conditioning without retraining the world action model.
- Helped evaluate the guidance with Fast-WAM and DreamZero. The shared system raised success from 64.4% to 70.0% on nine RoboTwin 2.0 tasks and from 69% to 75% in zero-shot simulation.

Shared-Compute Image Generation Pipeline Feb 2025 – May 2026
MIPAL, Seoul National University
Advised by Prof. Nojun Kwak and Prof. Jangho Kim (Kookmin University)

- Built a generation pipeline that shares backbone computation across samples, increasing DDPM throughput by 1.94x on CIFAR-10 with 0.29% additional parameters. Measured FID of 3.88 against the baseline's 3.41.
- Implemented training and evaluation for U-Net, DiT, and autoencoder backbones. Balanced throughput and sample fidelity by scheduling how many samples share computation at each generation step.

On-Device Threat Detection for Marine Safety Sept – Dec 2024
MIPAL, Seoul National University
Advised by Prof. Nojun Kwak

- Built a drone-image detection pipeline to identify threats near people, using YOLO for object detection and post-training quantization to prepare the model for on-device deployment.

Medical Image-and-Text Answer Generation Jan – June 2024
MIPAL, Seoul National University
Advised by Prof. Nojun Kwak

- Developed a multimodal answer-generation pipeline by fine-tuning and comparing MedVInT-Decoder, LLaVA, and LLaVA-Med on skin images and patient questions. The team ranked second in the MEDIQA-M3G English track, and post-challenge LLaVA reached BLEU 6.98.

Synthetic Vibration Data for Bearing Fault Detection Jan – Aug 2024
Hyperautomation Artificial Intelligence Lab, Seoul National University
Advised by Prof. Byeng Dong Youn

- Addressed scarce fault examples by generating abnormal vibration signals for journal-bearing fault detection. Received the Outstanding BS Thesis Presentation Award for this work.

Education

Seoul National University

Ph.D. in Intelligence and Information, in progress — Sept 2026 – Present
Advised by Prof. Nojun Kwak

M.S. in Engineering, Department of Intelligence and Information — Sept 2024 – Aug 2026
GPA 4.11/4.30. Advised by Prof. Nojun Kwak

B.S. in Mechanical Engineering, with an Interdisciplinary Major in AI — Mar 2019 – Aug 2024
GPA 3.86/4.30. Graduated cum laude

Selected Publications

- **ManiFusion**. First author. *NeurIPS 2026*, accepted. Multi-sample generation through shared backbone computation.
- **Policy-Aligned Predictive Verification**. First author. *Under review*, 2026. Adaptive execution of VLA action chunks.
- **Completion Aware Guidance for World Action Models**. Co-author. *NeurIPS Workshops 2026 – Robotics World Modeling*, accepted.
- **Complementary Future Token Pruning for World Action Models**. Co-author. *arXiv preprint*, 2026.
- **Recursive Harness Distillation across Agents for Robot Manipulation**. Co-author. *arXiv preprint*, 2026.
- **TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Dermatological Diagnosis**. Co-author. *NAACL 2024 – ClinicalNLP Workshop*.

Honors and Leadership

- Cum laude, Seoul National University, 2024.
- Outstanding BS Thesis Presentation Award, Mechanical Engineering, SNU, 2024.
- Bronze prize, Mechatronics Competition, Hyundai Infracore and SNU, 2023.
- South Korea National Delegation, Intel International Science and Engineering Fair (ISEF), May 2017.
- Captain, Engineering College Basketball Team, SNU, 2024.
- Sergeant and Squad Leader, Republic of Korea Army, Jan 2021 – July 2022.